PROJECT PROPOSAL

AR Campus Navigation for THM Friedberg

An augmented-reality wayfinding assistant that guides students and visitors across the THM campus in Friedberg.

Module	Augmented Reality - Summer Semester 2026
Supervisors	Prof. Dr.-Ing. Hartmut Weber · M.Sc. Severin Stahl
Project team	Tarik Billa · Sai Radhika Mitta
Framework	Google ARCore (Android)
Core sensors	GPS, compass / orientation sensors, camera

1. Project Description

This project is an Augmented Reality campus navigation application for the THM campus in Friedberg, developed with the ARCore framework on Android. Using the smartphone's GPS, compass and camera, the app first displays the user's live position relative to the campus buildings on a map, and then, in AR mode, overlays directional indicators and route guidance directly onto the camera view of the real path ahead. This gives first-semester students, applicants and visitors a concrete everyday advantage: they can quickly find the correct building and entrance without prior knowledge of the campus layout. In a later development stage the system is intended to be extended toward indoor guidance for locating specific floors and rooms.

2. Motivation and User Advantage

Finding the right building, entrance or lecture hall on an unfamiliar campus is a common and stressful problem, particularly for new students during their first days and for external visitors attending events or examinations. Conventional maps and static signage require the user to mentally translate a top-down plan into the real world, which is error-prone and slow. By anchoring guidance to the user's actual position and turning the live camera view into an intuitive follow-the-arrow experience, the application removes this translation step and makes orientation on campus noticeably faster, clearer and more confident.

3. System Architecture

The application follows a simple, robust pipeline. Three device sensors provide the raw input, a single processing stage derives the user's position together with the bearing and distance to each stored campus building, and the result is presented through two complementary navigation modes.

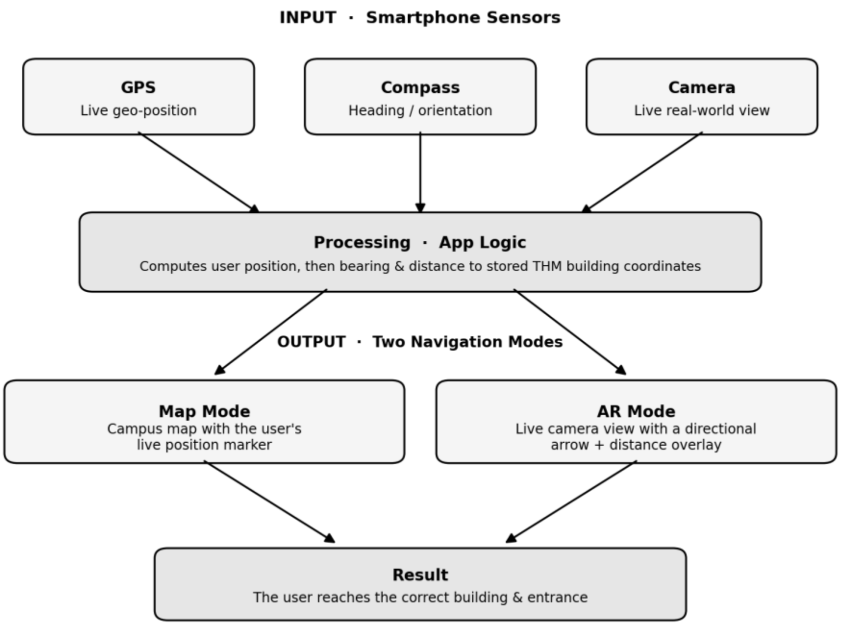

Figure 1: High-level system architecture of the AR campus navigation application.

4. Scope and Development Phases

To keep the project realistic within the available time, the work is divided into two committed phases and one optional outlook:

1. **Map mode –** display a campus map showing the user's GPS position together with the locations of the main THM Friedberg buildings as markers.
2. **AR camera mode –** open the camera and overlay directional indicators and distance toward the selected building directly onto the live view of the path ahead.
3. **Future extension (outlook) –** indoor navigation to locate specific floors and rooms once a building has been reached.

5. Expected Outcome

The intended deliverable is a working Android prototype that reliably guides a user between the main buildings of the THM Friedberg campus in both map and AR mode. Success is measured by the app correctly indicating the direction and distance to a chosen building from arbitrary positions on campus, so that a person unfamiliar with the site can reach the target entrance without additional help.

Note: Phases 1 and 2 constitute the core deliverable for the available project time. The indoor-navigation extension is documented as an outlook and is not part of the committed scope.